

A candidate proof of the Murty-Simon conjecture at order 30

Reviewer edition 1 - independent mathematical review pending

Paul Lenz research project; mathematical development and drafting with ChatGPT/Geeps

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Abstract

We present the current candidate argument that every simple diameter-2 edge-critical graph on 30 vertices has at most 225 edges, with equality exactly for $K(15,15)$. Published reductions leave a small set of maximum-degree cases; the difficult degree-16 and degree-17 frontiers are closed by exact finite kernels and integer Farkas certificates. The companion records the hostile assembly audit and replay boundary. Independent specialist review remains open.

Reviewer status. complete candidate; independent mathematical review OPEN. This reviewer edition is an editorial rendering of the canonical source proof listed below. It does not convert same-assistant checking into external independence, does not make a novelty or priority claim, and does not claim the unrestricted Murty-Simon conjecture unless the source proof itself proves such a statement.

Canonical claim. $e(G) \leq 225$, with equality exactly $K(15,15)$.

Canonical proof source. `project/reviews/n30/2026-09-09-candidate-v1/PROOF.md`. The source is reproduced in full below; substantive mathematical changes must be made in the canonical proof first and then rebuilt into this edition.

Murty-Simon at $n=30$: complete candidate proof

9 September 2026. Research directed by Paul Lenz; mathematical development, implementation and internal audit by ChatGPT/Geeps.

Status: complete candidate proof. Independent expert mathematical review, external computational reproduction and novelty assessment remain OPEN. This is not a proof of the unrestricted Murty-Simon conjecture.

1. Candidate statement

Let G be a finite simple diameter-two edge-critical graph on 30 vertices. The candidate statement is

$$e(G) \leq 225 = \text{floor}(30^{2/4}),$$

with equality if and only if

$$G \cong K(15,15).$$

Deleting an edge is understood to increase the diameter, with disconnected pairs assigned infinite distance.

2. Published upper reduction

Use Fan's strict upper bound in the form already used and cited in the $n=29$ candidate:

$$e(G) < n^2/4 + [n^2 - (81/5)n + 56]/320.$$

At $n=30$ the right-hand side is exactly

$$7247/32 = 226.46875.$$

Therefore a counterexample to $e(G) \leq 225$ can have only

$$m=226.$$

Equality at $m=225$ is handled separately; no edge is deleted and then assumed to leave an edge-critical graph.

A bipartite diameter-two graph is complete bipartite: a missing cross-part pair would have odd distance at least three. Hence any bipartite graph in the class has at most 225 edges, and equality on 30 vertices forces $K(15, 15)$.

It remains to treat non-bipartite graphs at $m=226$ and $m=225$.

3. Degree entry points

At $m=226$, the degree sum is 452, so

$$\Delta \geq 16.$$

At $m=225$, the degree sum is 450, so

$$\Delta \geq 15.$$

A universal vertex ($\Delta=29$) forces an edge-critical graph to be a star: any edge among the remaining vertices could otherwise be deleted while every pair remained within distance two. Thus $\Delta=29$ is sparse.

The proof now separates the degree values 15, 16, 17, and 18, . . . , 28.

4. Equality degree $\Delta=15$

This degree occurs only at $m=225$, because $30 \cdot 15/2 = 225$. Thus every vertex is degree 15.

Assume first that G is non-bipartite. At this density, the published dominating-edge result of Dailly, Foucaud and Hansberg used elsewhere in the project excludes a dominating edge (their non-bipartite dominating-edge upper bound is below 225, apart from the order-six exception).

Every critical edge has a direct or two-step witness. Briefly: delete a critical edge xy and choose a pair whose distance becomes greater than two. In the original graph, every path of length at most two between that pair uses xy . If the witness pair is adjacent it is a direct witness with no common neighbour; if nonadjacent it has exactly one common neighbour and its unique two-edge path uses xy .

For either witness type in a 30-vertex graph with no dominating edge,

$$d(u) + d(v) \leq 29.$$

For a two-step witness, the two neighbourhoods have intersection one and their union lies among the other 28 vertices, giving degree sum at most 29. For a direct witness, the neighbourhoods are disjoint; degree sum 30 would make their union all vertices and the witness edge dominating.

But in a 15-regular graph every pair has degree sum 30. Hence no witness can exist, contradicting edge-criticality.

Therefore the $\Delta=15$, $m=225$ graph cannot be non-bipartite. The bipartite observation from Section 2 then forces

$$G = K(15, 15).$$

Conversely, $K(15, 15)$ is diameter-two edge-critical: deleting a cross edge makes its endpoints distance three.

Thus the equality graph is uniquely determined once the $\Delta \geq 16$ equality scopes are excluded.

5. Common graph-to-demand bridge for $\Delta=16$ and 17

For the remaining dense scopes put $H = \text{complement}(G)$, choose a minimum-degree vertex v in H , and set

$$\begin{aligned} A &= N_H(v), \\ B &= V(H) \setminus N_H[v], \\ a &= |A| = n-1-\Delta, \\ b &= |B| = \Delta. \end{aligned}$$

Let $C = H[A]$, $F = \text{complement}(C)$ on A , and define the selected/residual cross-edge system as follows.

For every missing unordered pair uw of $H[B]$, adding uw corresponds to deleting a critical G -edge. The enlarged complement acquires a new adjacent total-dominating pair. It cannot be $\{u, w\}$, because both miss v ; therefore, after interchanging u, w if necessary, there is an existing cross-edge ui with i in A such that

$$N_H(u) \cup N_H(i) = V(H) \setminus \{w\}.$$

Write $ui \rightarrow w$, choose one such selected cross-edge for each missing B -pair, and call all other existing A - B edges residual. The source and unique exception recover the missing B -pair, so the selected assignment is injective.

Let ρ_u, R_i be residual row/column degrees, $r = \sum \rho_u = \sum R_i$, $d_i = \deg_F(i)$, and

$$\begin{aligned} s_i &= \max(0, d_i - R_i), \\ S &= \sum_i s_i, \\ t &= m - b(n-b). \end{aligned}$$

The exact ledger and minimum-complement-degree condition give

$$\begin{aligned} e(F) &= r + t, \\ S &\geq r + 2t. \end{aligned}$$

For $t > 0$, every B -source is residual-active: $\rho_u \geq 1$.

For a selected source of label i , the selected-edge forcing gives

$$s_i \leq \rho_u.$$

Charging chosen selected incidences then yields the universal necessary inequality

$$r - b \geq \sum_i s_i(s_i - 1) / (a - s_i),$$

and hence

$$\sum_i s_i(a + 1 - 2s_i) / (a - s_i) \geq b + 2t. \quad (5.1)$$

For thresholds $h \geq 1$, define

$$\begin{aligned} I_h &= \{i : s_i \geq h\}, \\ W_h &= \sum_{i \in I_h} s_i, \end{aligned}$$

$Z_h = \{u : \text{rho}_u \geq h\},$
 $z_h = |Z_h|.$

The exact source/supplement capacity argument gives

$$2W_h \leq z_h^2 - z_h + h(h+1). \quad (5.2)$$

The complete standalone derivations are preserved in the n=29 bridge package, especially GRAPH_TO_MODEL_BRIDGE and THRESHOLD_CAPACITY_LEMMA.md. The n=30 parameterization audits check that no n=29-specific constant is retained when these statements are instantiated below.

6. Delta=17 closes before residual-row enumeration

Here

$a=12,$
 $b=17.$

The two dense scopes have

$m=226 \rightarrow t=5,$
 $m=225 \rightarrow t=4.$

6.1 m=226

Complete enumeration of the nondecreasing integer demand profiles satisfying (5.1) gives exactly 250 profiles.

For each profile, charging gives an exact upper bound $r_{\max}$. Residual activity then implies

$$z_h \leq \text{floor}((r_{\max} - b) / (h - 1))$$

for $h \geq 2$. Substitution into (5.2) rejects every one of the 250 profiles by exact integer arithmetic. The smallest strict threshold margin is 2.

Thus $\Delta=17, m=226$ is impossible.

6.2 m=225

The complete charging domain has 1,155 profiles.

Exact threshold/source-count inequalities reject

1,070 by threshold capacity,
 67 by source count,

leaving 18 profiles.

For the k largest demands, let D_k be their total. A residual-degree- j source has at most $a-j$ selected slots and can serve only demands at most j , so its contribution to that prefix is at most

$$A[k, j] = \min(a - j, \#\{\text{top-}k \text{ labels with demand} \leq j\}).$$

Every graph-realizable source-count vector n_j satisfies

$$\begin{aligned} D_k &\leq \sum_j A[k, j] n_j, \\ \sum_j n_j &= 17, \\ \sum_j (j-1)n_j &\leq r_{\max} - 17. \end{aligned}$$

All 18 remaining profiles have exact integer dual contradictions to this necessary Hall system. The weakest exact dual margin is 3.

The clean replay 34292054922 completed successfully: the complete demand frontier was regenerated, the same 18 post-threshold profiles were recovered, and both regenerated and committed dual certificates were verified by a standard-library exact checker.

Hence $\Delta=17$, $m=225$ is impossible.

7. $\Delta=16$: parameterized trusted kernel

Here

$a=13$,
 $b=16$,

and

$m=226 \rightarrow t=2$,
 $m=225 \rightarrow t=1$.

Both have $t>0$, so residual activity applies.

7.1 Isolated-C exclusion

The parameteric isolated-C lemma says that if $C=H[A]$ has an isolated vertex then necessarily

$$b \leq a-1-t. \quad (7.1)$$

Its expanded proof is preserved as ISOLATED_C_PARAMETERIC_LEMMA.md in the n=30 reconnaissance directory. At $(a, b) = (13, 16)$, (7.1) would require

$16 \leq 10$ at $t=2$,
 $16 \leq 11$ at $t=1$,

both impossible. Therefore

$\Delta(C) \geq 1$,
 $d_i \leq 11$,
 $e(C) \geq 7$.

Using $e(C) + r = C(13, 2) - t$, the safe residual-total bounds are

$r \leq 69$ at $m=226$,
 $r \leq 70$ at $m=225$.

No order-specific graph lemma beyond the parameterized bridge is added.

7.2 Demand preparation and residual rows

The exact strengthened preparation uses only charging, residual activity, (7.1), threshold capacity, and exact source-capacity Hall dual lower bounds on r .

At $m=226$:

48,046 charging-domain profiles
 $\rightarrow$ 2,590 retained demand profiles.

At $m=225$:

67,050 charging-domain profiles
 $\rightarrow$ 5,379 retained demand profiles.

The clean residual-row scanner enumerates every sorted length-16 residual row in the certified total interval and uses only necessary largest-demand-prefix Hall bounds and monotone supplement-cap refinement.

Clean workflow run 34286806474 gives:

m	raw residual states	initial Hall rejects	refinement rejects	row survivors
226	50,690,620	50,561,364	93,726	35,530
225	158,314,695	157,894,303	269,496	150,896

These are arithmetic necessary-condition states, not graphs.

7.3 Exact row-threshold screen

Once a concrete residual row is known, (5.2) can use the exact value

$$z_h = |\{u : \rho_u \geq h\}|.$$

The exact row-threshold workflow 34287440190 reduces the frontiers to

```
m=226: 9 rows,
m=225: 272 rows.
```

7.4 Fresh corrected cumulative-threshold/source-flow model

The remaining 281 rows are tested by a fresh $n=30$ implementation specialized directly to $(a, b) = (13, 16)$. It does not import the historical $n=29$ threshold builders.

The grouped normalization uses the corrected per-label identity

$$\sum_k n_k Z = \sum_h T_h,$$

with no extra label-group multiplicity. The same-group ordered-pair capacity is deliberately relaxed, not strengthened, so it can create false survivors but not false exclusions.

A numerical LP result is never itself an exclusion. HiGHS is used only to propose a Farkas ray; integer multipliers are then reconstructed and accepted only if direct exact arithmetic verifies coefficientwise non-negativity and a strictly negative combined right-hand side.

Clean final workflow 34287739057 completed both jobs successfully:

```
m=226: 9/9 exact Farkas rejections, zero survivors,
m=225: 272/272 exact Farkas rejections, zero survivors.
```

Thus both $\Delta=16$ dense scopes are impossible within the audited bridge framework.

8. $\Delta=18, \dots, 28$

For every $\Delta=18, \dots, 28$ at both $m=225$ and $m=226$, instantiate (5.1) with

$$\begin{aligned} a &= 29 - \Delta, \\ b &= \Delta, \\ t &= m - \Delta(30 - \Delta). \end{aligned}$$

The complete nondecreasing integer demand domain is empty in every scope: there is no profile $0 \leq s_i \leq a-1$ satisfying the charging inequality.

This is reproduced by the standard-library checker `check_outer.py` in this candidate directory.

Hence every $\Delta=18, \dots, 28$ scope is impossible. $\Delta=29$ is the universal-vertex/star case already excluded as sparse.

9. Assembly

At $m=226$:

- $\Delta \leq 15$ is impossible by degree sum;
- $\Delta=16$ is excluded by the parameterized trusted kernel;
- $\Delta=17$ is excluded by threshold capacity;
- $\Delta=18, \dots, 28$ are excluded by charging;
- $\Delta=29$ gives a star.

Therefore no 226-edge counterexample exists, and Fan's strict bound gives

$e(G) \leq 225$.

At $m=225$:

- $\Delta \leq 14$ is impossible by degree sum;
- $\Delta=15$ forces the unique equality graph $K(15, 15)$;
- $\Delta=16$ is excluded by the parameterized trusted kernel;
- $\Delta=17$ is excluded by threshold/source-count plus 18 exact Hall duals;
- $\Delta=18, \dots, 28$ are excluded by charging;
- $\Delta=29$ gives a star.

Thus the complete candidate statement is

$e(G) \leq 225$,
with equality exactly $K(15, 15)$.

10. Trust boundary

This remains a candidate theorem. The principal remaining risk is the correctness of the universal graph-to-demand bridge, not the arithmetic replay.

Highest-value external review targets are:

1. complement/quasi-edge construction and selected-edge injection;
2. residual activity for $t > 0$;
3. selected-source demand $s_i \leq \rho_u$;
4. charging inequality;
5. threshold-capacity lemma;
6. the parameteric isolated-C lemma;
7. the $n=30$ $\Delta=16$ grouped LP normalization and graph-to-relaxation embedding;
8. exact Farkas checker semantics;
9. the use and exact hypotheses of the cited published Fan and dominating-edge reductions.

Any counterexample to a universal bridge lemma overrides every green workflow.